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Department of Artificial Intelligence and Machine Learning

Date: 16/12/2025	Test - II	Max. Marks:50
Semester: V	UG	Duration: $1\frac{1}{2}$Hrs
Course Title: Machine Learning Operations [MLOps]		Course Code:AI254TA

Sl no	QUESTIONS – PART B	M	BT	CO
1a.	Discuss the need of reproducibility and Auditability in building MLOps pipeline.	4	2	5
1b.	Explain how automated testing pipelines CI/CD reduce deployment failures in ML projects. Give example	6	3	4
2a.	What are the two different types of model deployment approaches used in MLOps? Explain with suitable examples	4	2	2
2b.	An autonomous vehicle company uses ML models for object detection. Hackers attempt to manipulate street signs to mislead the vehicle's perception system. Explain how machine learning security and risk mitigation measures can safeguard production systems in such safety-critical applications.	6	4	3
3a.	Elaborate the key considerations when preparing machine learning models for production deployment.	6	4	3
3b.	"For batch scoring, the scaling deployments can be more complex". Analyse the statement and discuss the two types of strategies to distribute the computation.	4	2	3
4a.	A healthcare analytics company uses an ML model to predict early signs of cardiac risk based on live patient vitals. To upgrade to a new model version without interrupting critical monitoring, the team wants to use a Blue-Green deployment approach. Explain how Blue-Green deployment ensures safe and zero-downtime release of the updated model. Enumerate the potential risks if real-time monitoring traffic suddenly switches to the new model and provide the steps mitigate the risks.	5	4	3
4b.	Describe the significance of containerization technologies in MLOps and evaluate the challenges faced while working with Docker environments.	5	4	5
5	A smart-city project deploys an ML-based traffic prediction system that uses live sensor data, historical traffic flow, and weather conditions to optimize signal timings and reduce congestion. The model is developed in a controlled environment and then pushed to production using MLOps practices such as automated data pipelines, model validation, deployment workflows, and real-time monitoring. Examine the risks involved when migrating this ML system from development to production, focusing on data reliability issues, model drift, integration with IoT devices, latency challenges, and security threats. Justify the importance of selecting a suitable runtime environment to ensure low-latency, scalable, and reproducible ML operations, Justify your solution.	10	4	3